Experiment-7

Write A Python Program To Implement BFS

Aim

To write and execute a Python program to implement Breadth-First Search (BFS) for traversing a graph level by level.

Algorithm

1. Start the program.
2. Represent the graph using an adjacency list.
3. Select the starting node.
4. Initialize an empty set to store visited nodes.
5. Create a queue and insert the starting node.
6. Check whether the queue is empty.
7. Remove the first node from the queue.
8. If the node is not visited, mark it as visited.
9. Display the visited node.
10. Add all its unvisited neighboring nodes to the queue.
11. Repeat Steps 6–10 until the queue becomes empty.
12. Stop the program after completing the traversal.

Code

ex-7.py - D:/ex-7.py (3.10.0)

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```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    print("BFS Traversal:")

    while queue:
        node = queue.popleft()

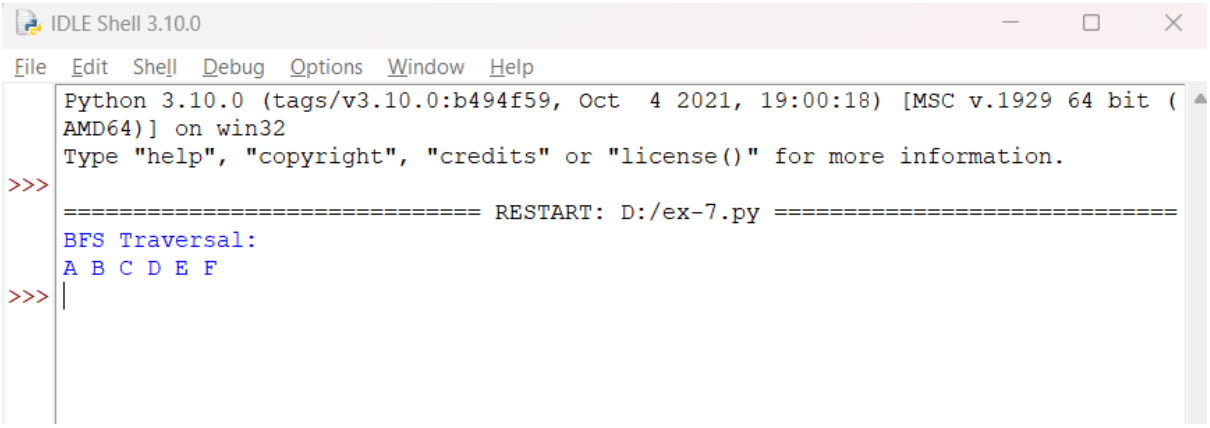
        if node not in visited:
            print(node, end=" ")
            visited.add(node)

            for neighbor in graph[node]:
                if neighbor not in visited:
                    queue.append(neighbor)


# Graph representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

# Start BFS from node A
bfs(graph, 'A')
```

Output



Result

The Breadth-First Search (BFS) algorithm was successfully implemented . The program traverses the graph starting from node A and produces the output:

A B C D E F